

## Article

## New insights to compare and choose TKTD models for survival based on an inter-laboratory study for *Lymnaea stagnalis* exposed to Cd

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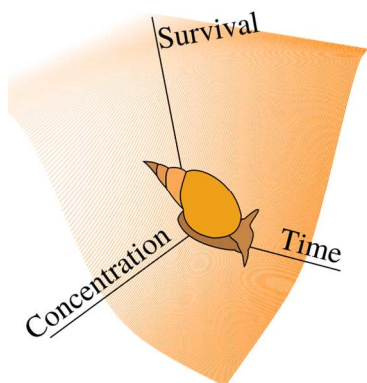
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10 **TOC-Art**



## ABSTRACT

Toxicokinetic-toxicodynamic (TKTD) models, as the General Unified Threshold model of Survival (GUTS), provide a consistent process-based framework compared to classical dose-response models to analyze both time and concentration-dependent data sets. However, the extent to which GUTS models (Stochastic Death (SD) and Individual Tolerance (IT)) lead to a better fitting than classical dose-response model at a given target time (TT) has poorly been investigated. Our paper highlights that GUTS estimates are generally more conservative and have a reduced uncertainty through smaller credible intervals for the studied data sets than classical TT approaches. Also, GUTS models enable estimating any  $x\%$  lethal concentration at any time ( $LC_{x,t}$ ), and provide biological information on the internal processes occurring during the experiments. While both GUTS-SD and GUTS-IT models outcompete classical TT approaches, choosing one preferentially to the other is still challenging. Indeed, the estimates of survival rate over time and  $LC_{x,t}$  are very close between both models, but our study also points out that the joint posterior distributions of SD model parameters are sometimes bimodal, while two parameters of the IT model seems strongly correlated. Therefore, the selection between these two models has to be supported by the experimental design and the biological objectives, and this paper provides some insights to drive this choice.

## KEYWORDS

GUTS models, classical dose-response models, constant exposure, aquatic toxicity,  $x\%$  lethal concentration

## INTRODUCTION

New chemicals are created every day to be used among others as industrial products, pesticides, pharmaceuticals and so on, for which safety testing may be required in most parts of the world [1]. In the European Union, the REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) regulation has been adopted in 2007 in order to “improve the protection of human health and the environment from the risks that can be posed by chemicals” [2]. It demands to companies the identification and management of risks linked with every substance in the context of their usages. Since 1981, the Organization for Economic Cooperation and Development (OECD), an intergovernmental organization, develops the OECD Guidelines for the Testing of Chemicals, which consist in internationally agreed methods for the testing of chemicals [1]. These guidelines are generally matching the expectations of the REACH and other regulations, and are consequently updated and enhanced frequently. For instance, reproductive toxicity tests methods on molluscs are being developed by the OECD since 2011 to support chemical risk assessment [3,4]. Corresponding test guidelines have been published in 2016 [5–7], which include statistical modelling approaches for data analysis.

These statistical approaches are derived from the 2006-published OECD guideline, which provides practical guidance on how to analyse the data from toxicity tests [8]. It specifies that dose-response modelling is well adapted to estimate the  $EC_x$  of a chemical substance (the effective concentration that causes x% of effect, being an  $LC_x$  if the response is death), as well as its uncertainty. When the dose-response modelling relates to a given exposure duration, we will call this approach a target time (TT) analysis. Several dose-response models exist and are frequently used to analyse collected data from toxicity tests [8].

As TT analyses do not take into account both time and concentration simultaneously, using them to model data sets depending on these two variables may raise some issues. For instance, in TT analyses, only data at a given TT are used, so that parameters strongly depend on the chosen TT. In addition, the large part of ignored observations may lead to a lower accuracy on parameter estimates. Toxicokinetic-toxicodynamic (TKTD) models, which simulate the time-course of processes leading to toxic effects, may then represent a promising alternative. TKTD models consist of two parts. First, the toxicokinetic part (TK) translates the external concentration of the chemical substance in the medium into an internal concentration within the exposed organism over time. Then, the toxicodynamic part (TD) links this internal concentration to the effects on life-history traits over time [9]. A very large number of TKTD models exist to describe different effects of chemical substances on life history traits. The General Unified Threshold model for Survival (GUTS) has been suggested by Jager et al. (2011), making GUTS a standardized framework unifying TKTD models for survival [9]. Jager's study showed that a lot of existing survival models can be derived from GUTS as special cases with reduced versions of model equations [9]. Among these special cases, Stochastic Death (SD) and Individual Tolerance (IT) models are probably – nowadays – the most used ones. Stochastic Death models assume that all individuals are identically sensitive to the tested chemical substance above a certain internal threshold concentration and that mortality is a stochastic process once this threshold is reached. On the contrary, Individual Tolerance models are based on the Critical Body Residues (CBR) approach, which assumes that each organism dies as soon as its internal concentration reaches its own internal threshold. The individual internal threshold concentration is then assumed to follow a probability distribution among the organisms [9].

In the present study, we revisited the data sets from the two ring-tests studied by Ducrot et al. in 2014 [3] and Charles et al. in 2016 [5]. These ring-tests were performed in the scope of the consolidation of the OECD Test Guideline for the assessment of chemical effects on *Lymnaea stagnalis* (the Great Pond Snail) reproduction [3,5,6]. The present study focused however only on the survival of this hermaphrodite freshwater snail when exposed to cadmium (Cd). Although reproduction is not under focus in this paper, the data from the ring-tests were chosen as a basis for our work because they provide a unique collection of survival data over the long-term: this is ideal for our purpose to compare different modelling tools in their ability to address the influence of the interaction between time and concentration on survival during laboratory toxicity tests.

The first objective of our paper was indeed to model the time-course of survival of *L. stagnalis* exposed to Cd, with both GUTS-SD and GUTS-IT approaches, and to compare the performance of the two models. Our second objective was to compare the results of the TKTD approach with the results of the more classical analysis at TT.

## MATERIALS AND METHODS

### *Principle of the toxicity test*

To evaluate the impact of Cd on *L. stagnalis* (simultaneous hermaphrodite), 6 replicates of 5 reproducing adults (that is 30 snails per treatment) were exposed to a range of fixed concentrations of Cd, and 6 additional replicates of 5 snails used as controls. Prior to the tests, snails were sampled from a parasite-free cohort, checked for identical size ( $27 \pm 2$  mm), shell integrity and reproduction ability [3,5]. As regards Cd, from 5 to 7 concentrations per laboratory were tested, ranging from 16 to 486  $\mu\text{g.L}^{-1}$  (average measured concentrations). The concentration, the temperature ( $20^\circ\text{C} \pm 1^\circ\text{C}$ ) and the dissolved oxygen concentration ( $[O_2] > \text{air saturation value}$ ) were maintained constant over 56 days long, during which the reproduction and

the mortality of snails were monitored twice a week [3]. Once counted and recorded, dead snails were removed from the test vessels.

This experimental protocol was applied twice: by seven European laboratories between 2011 and 2013 (prevalidation ring-test) from which two have been removed because of failure in the protocol (all individual dying whatever the concentration; or non-significant effects were recorded on reproduction, the life-history trait of interest in the study) thus giving five data sets (hereafter denoted Labs 01, 04, 07, 14 and 15) ; and by thirteen laboratories between 2013 and 2014 (validation ring-test), among which six (not involved in the prevalidation ring-test) were testing Cd (hereafter denoted Labs 02, 05, 06, 10, 11 and 13). Consequently, we used eleven data sets, five from the prevalidation ring-test, and six from the validation ring-test. The experience of all these laboratories with mollusk testing varied a lot, from inexperienced to expert ones. For more information on the experimental protocol, see [3,5].

#### *Significance of the effects of cadmium on survival*

As done for reproduction [5], we first checked if effects of Cd on survival were significant (in comparison to controls) and concentration dependent. This was performed using the Jonckheere-Terpstra hypothesis test, a nonparametric test determining the statistical significance of trend between ranked variables [10]. This test was run under the R software [11], with the ‘clinfun’ R-package and the ‘jonckheere.test’ function [12]. Control samples were used as reference and the number of iterations was fixed to  $10^6$ . Results from these analyses showed that the survival rate significantly decreased with increasing Cd concentrations (Jonckheere-Terpstra p-values at day 56  $< 0.05$  as shown in Table S1 in Supporting Information) for all laboratories.

#### *Modelling of survival data*

### Principle of GUTS models

In the following, we detail the mathematical equations of the GUTS approach describing the survival rate of organisms exposed to a constant concentration of Cd over time. During the chronic toxicity tests we studied, concentrations  $c_i$  ( $i$  varying between 1 and 5 or 1 and 7) of Cd were indeed held constant and the mortality was recorded twice a week. Moreover, no measurement of internal concentration was performed during the tests, meaning that the scaled internal concentration is a latent variable as described by the toxicokinetics model part detailed hereafter by equation (1).

If we note  $N_{i,k}$  the number of survivors at time  $t_k$  (with  $t_0 < t_1 < \dots < t_m$  and  $t_0 = 0$ ) and concentration  $c_i$ , the data sets are composed of triplets  $D = \{(c_i, t_k, N_{i,k})\}_{i,k}$  of observations. As only the external concentration  $c$  is available in our data sets, the TK part of GUTS translates first the external concentration  $c$  into a scaled internal concentration  $c_{int}$ :

$$\frac{dc_{int}(t)}{dt} = k_d(c - c_{int}(t)) \quad (1)$$

$k_d$  being the dominant rate constant, corresponding to the slowest compensating process (damage repair or elimination of the toxicant) dominating the overall dynamics of toxicity [9].

The number of survivors  $N_{i,k}$  at time  $t_k$  given the number of survivors  $N_{i,k-1}$  at time  $t_{k-1}$  is assumed to follow a conditional binomial distribution characterized by the number  $N_{i,k-1}$  of organisms. The conditional probability for an organism to survive between times  $t_{k-1}$  and  $t_k$  is given by [13,14]:

$$N_{i,k} \sim B(N_{i,k-1}, \frac{S_i(t_k)}{S_i(t_{k-1})}), k \geq 1 \quad (2)$$

$S_i(t_k)$  is the probability to survive until time  $t_k$  under external concentration  $c_i$  (with  $S_i(t_0) = 1$ ). The expression of  $S(t)$  depends on the model and will consequently be detailed further for GUTS-SD and GUTS-IT.

#### *Specific assumptions of GUTS-SD*

The GUTS Stochastic Death model (GUTS-SD) supposes, as mentioned in the introduction, that all the organisms have the same internal threshold concentration (denoted  $z$  and also called NEC for No-Effect-Concentration), and that, once exceeded, the instantaneous probability to die increases linearly with the internal concentration  $c_{int}$  [9]. At time  $t_0$  in our toxicity tests, the internal concentration of Cd is assumed to be null for every organism at every external concentration  $c$  in the water, and so no death occurred. In addition, if  $c$  is above  $z$ , we need to define time  $t_z$  from which the internal concentration,  $c_{int}(t_z)$ , is equal to threshold  $z$ , leading to death events among the observed organisms [9]. Time  $t_z$  is then defined as dependent on threshold  $z$ :

$$t_z = -\frac{1}{k_d} \ln\left(1 - \frac{z}{c}\right) \quad (3)$$

Once time  $t_z$  is defined, it is possible to write the expression of  $S(t)$ , the probability to survive until time  $t$  under concentration  $c$ , as in [9] by:

$$S(t) = \exp\left(\int_0^t -h(\tau) d\tau\right) \quad (4)$$

With  $h$  being the instantaneous probability to die (also called hazard rate) defined as follows:

$$h(\tau) = k_k \max(0, c_{int}(\tau) - z) + h_b \quad (5)$$

163 with  $k_k$  being the killing rate,  $z$  the threshold concentration for survival of all the organisms and  
 164  $h_b$  the background hazard rate (i.e., the instantaneous mortality rate in the water control).  
 165 Equation (5) translates the fact that the instantaneous probability to die increases linearly once  
 166  $c_{int}(t)$  is above  $z$ .

167 Based on [13], if internal concentration  $c_{int}$  is below  $z$ , or if time  $t$  is below  $t_z$ , the survival  
 168 function is consequently expressed as:

$$169 \quad S(t) = \exp(-h_b t) \quad (6)$$

170 Otherwise:

$$171 \quad S(t) = \exp(-h_b t - k_k(c - z)(t - t_z) - \frac{k_k c}{k_d}(\exp(-k_d t) - \exp(-k_d t_z))) \quad (7)$$

172 Based on equations (6) and (7), it is possible to determine any  $LC_x$  at any time  $t$  whatever  $x$   
 173 (once the model parameters are estimated), as being the concentration  $c$  for which

$$174 \quad S(t) = (1 - x/100)S(t_0) \quad (8)$$

175 We set  $x = 50$  to get the  $LC_{50}$  at any time  $t$ .

#### 176 *Specific assumptions of GUTS-IT*

177 For the GUTS-IT model, we assume that the threshold concentration is distributed among the  
 178 organisms. Here, we assume the distribution to be log-logistic with a median  $\alpha$  and a shape  $\beta$ .  
 179 The probability to survive until time  $t$  and concentration  $c$  can then be expressed as follows:

$$180 \quad S(t) = \exp(-h_b t) \left(1 - \frac{1}{1 + \left(\frac{c(1 - \exp(-k_d t))}{\alpha}\right)^{-\beta}}\right) \quad (9)$$

181 As for GUTS-SD, this expression enables to determine any  $LC_x$  at any time  $t$  whatever  $x$ .

*Implementation of the GUTS models*

To implement the GUTS models (SD and IT), we chose the R-package ‘morse’ [15] that allows the simultaneous use of JAGS and R software thanks to the R-package ‘rjags’ [16]. Package ‘morse’ consequently enables the implementation of Bayesian inference.

There are multiple ways to implement TKTD models, either using a frequentist approach, see for instance Albert et al., 2016 [17]; or Bayesian inference based on Bayes rule, see for example Delignette-Muller et al., 2017 [13] who illustrated a higher robustness of the Bayesian framework compared to the frequentist one in the case of sparse data sets. In addition, the use of Bayesian inference has already provided promising results in ecotoxicology [18-23].

Bayesian inference consists in fitting a probability model to a data set and results in probability distributions of the model parameters as well as of unobserved quantities [24]. Bayesian inference is based on the use of Monte Carlo Markov Chains (MCMC) simulations to infer the parameter estimates of every implemented model. This statistical method uses observations to update what is already known on the parameters (priors), in order to provide their joint posterior distributions [24]. For each one of the three models and each data set, three independent MCMC chains were run in parallel. The number of iterations and the thinning of these chains were determined by using the Raftery and Lewis method [25] as implemented in the ‘rafter.diag’ function of the R-package ‘rjags’. We finally checked the chain convergence by using the Gelman statistics [24].

For the GUTS-SD model, the choice of the four prior distributions for parameters  $h_b$ ,  $k_d$ ,  $k_k$  and  $z$ , is based on [13], which favours weakly informative priors in a log scale based on simple assumptions such as the threshold for effects is expected to be in the range of the tested

concentrations. So, whatever parameter  $\theta$ , a realistic range of values was defined between a lower value  $\theta_{min}$  and a maximum value  $\theta_{max}$ . From these two extreme values, a log-normal prior distribution was defined as follows:

$$\log_{10}\theta \sim N\left(\frac{\log_{10}(\theta_{min}) + \log_{10}(\theta_{max})}{2}, \frac{\log_{10}(\theta_{max}) - \log_{10}(\theta_{min})}{4}\right) \quad (10)$$

For the GUTS-IT model, concerning the three parameters  $h_b$ ,  $k_d$  and  $\alpha$ , the chosen prior distributions were similar to the ones used for the GUTS-SD analysis (equation (10)). The prior distribution of  $\beta$  is however defined differently, by a non-informative log-uniform distribution between -2 and 2.

In order to avoid any confusion, the parameters defining each one of the two models are gathered together in Table 1.

#### *Goodness-of-fit*

Goodness-of-fit of both GUTS models (SD and IT) was assessed by computing the Deviance Information Criterion (DIC) and performing a cross-validation. The DIC is a classical Bayesian measure of predictive accuracy based on posterior estimates. The cross-validation is more robust [24] whereas less used because time-consuming. In cross-validation, models are fitted on a "training" data set, and the resulting parameter estimates are used to calculate the predictions for the other data sets (so-called "validation" data sets). Each of the eleven laboratories was successively used for training and validation, as well as the pool of data from both the pre-validation and the validation ring-tests, leading to a total of thirteen data sets for the cross-validation. For each posterior estimate of the thirteen data sets, we also calculated the prediction for the entire pool of pre-validation and validation data. Then, the goodness-of-fit of all kinds of predictions was quantified from the percentage of observed data lying within the 95% predicted

credible interval. A prediction is considered as good when around 95% of the observations lie within the 95% predicted credible intervals. The results of the cross-validation are summarized in a 14x13 matrix of these percentages.

### *Target time analysis*

This classical analysis of survival data at a given target time (denoted  $TT$ , usually fixed at the latest observed time point) is based on the log-logistic model [14]. In this modelling approach, the data sets are composed of triplets of observations  $D = \{(c_i, N_{i,init}, N_{i,TT})\}_i$ , where  $c_i$  is the concentration in the media ( $i$  varying between 1 and 5 or 1 and 7),  $N_{i,init}$  the initial number of individuals at concentration  $c_i$  and  $N_{i,TT}$  the number of survivors at  $TT$  and concentration  $c_i$ . This model supposes that the deaths of two organisms are two independent events and that the survival rate after  $t$  days is a function  $f$  of the concentration. Consequently, the number  $N_{i,TT}$  of survivors at  $TT$  follows a binomial distribution [15]:

$$N_{i,TT} \sim B(N_{i,init}, f(c_i)) \quad (11)$$

$f(c)$  being the mean survival rate at  $TT$  and concentration  $c$ , defined as [14]:

$$f(c) = \frac{d}{1 + \left(\frac{c}{e}\right)^b} \quad (12)$$

where  $d$ ,  $e$  and  $b$  are positive parameters corresponding to the survival rate in the control (parameter  $d$ ), the  $LC_{50}$  (parameter  $e$ ) and a value related to the effect intensity of the chemical substance (parameter  $b$ ).

By analogy with the GUTS models, and as provided within the R-package ‘morse’, prior distributions of  $e$  and  $b$  were defined as:

$$\log_{10}e \sim N\left(\frac{\log_{10}(\min_i(c_i)) + \log_{10}(\max_i(c_i))}{2}, \frac{\log_{10}(\max_i(c_i)) - \log_{10}(\min_i(c_i))}{4}\right) \quad (13)$$

$$\text{and } \log_{10}b \sim U(-2, 2) \quad (14)$$

As regards the prior distribution of parameter  $d$ , it is assumed to be uniform between 0 and 1.

## RESULTS

### *GUTS-SD and GUTS-IT analyses*

We compared the observed and the estimated number of survivors provided as medians and 95% credible intervals for GUTS-SD and GUTS-IT models. This comparison is represented on Figure 1 for Lab.14 and on Figure S1 in Supporting Information for the other laboratories of the two ring-tests and both models. The results obtained for Lab.14 are indeed quite representative of the overall results of all laboratories and of both models. This laboratory will consequently be used to illustrate all outputs hereafter. Whatever the goodness-of-fit of those graphs, all of them reveal very similar patterns for both SD and IT models. In addition, we compared the observed and estimated values by plotting the Posterior Predictive Check (PPC) [15, 26], as shown on Figure S2 in Supporting Information. Here again, GUTS-SD and GUTS-IT appear very similar, as also pointed out by the average percentage of observed points in the 95% credible intervals of the predictions which are 96.32% for both GUTS-SD and GUTS-IT models for Lab. 14. As illustrated by the diagonal of Figure 2, those percentages remain almost the same whatever the laboratory for both GUTS-SD and GUTS-IT models.

The goodness-of-fit measured with the Deviance Information Criterion (DIC) indicates that the GUTS-SD model is slightly better (with a mean DIC at 199.94 and a standard deviation at 46.57) than the GUTS-IT model (with a mean DIC at 211.54 and a standard deviation at 52.65). *A contrario*, based on the cross-validation (Figure 2), pairwise comparisons between all

laboratories gives mean percentages at 91.55% for the GUTS-SD model and at 92.08% for the GUTS-IT model. This suggests a slightly better fit with the GUTS-IT model. The cross-validation illustrates that predictions from both GUTS-SD and the GUTS-IT models based on parameter estimates from any laboratory or any combination of the laboratories (read columns in Figure 2) are reliable whatever the laboratory for which these predictions are made for.

Implementing MCMC simulations to infer all model parameter estimates finally generated a large sample of the joint posterior distribution. The median value of the four parameter estimates and their 95% credible intervals (defined as the range between the 2.5% and 97.5% quantiles) are presented in Table S2 for all laboratories and both models. The joint posterior distribution can also be projected in each plane of parameter pairs in order to visualize the correlation between parameters of each laboratory as shown on Figure S3 in Supporting Information. Finally, we noticed that all inference outputs for the GUTS-SD and the GUTS-IT models were very close whatever the laboratory.

We compared common parameters of GUTS-SD and GUTS-IT ( $k_d$ ,  $h_b$  and  $z$  with  $\alpha$ ) for all laboratories (see Figure S5 in Supporting Information). For each laboratory, the dominant rate constant ( $k_d$ ) is higher in GUTS-SD than in GUTS-IT, with a strong difference for some data sets (Labs 01, 04, 15 and 11). No typical pattern appears when comparing background mortality  $h_b$  for both models. The no-effect concentration threshold ( $z$ ) and the median of the threshold distribution ( $\alpha$ ) show a high similarity between both models, except for Lab.10 that exhibits a large 95% credible interval for  $z$  of GUTS-SD compared to  $\alpha$  of GUTS-IT.

### ***Comparison of GUTS models and target time analysis***

As explained in the Materials and Methods part, it is possible for both GUTS models to estimate the  $LC_{50}$  at any exposure duration  $t$ , as being the concentration for which the probability to survive at time  $t$  is 0.5.  $LC_{50}$  values are also estimated with a TT analysis, as it is exactly parameter  $e$  (equation (12)).

The  $LC_{50}$  estimates (median and 95% credible intervals) under the three approaches are represented on Figure 3 for Lab.14 and on Figure S4 in Supporting Information for the other laboratories (boxplots represent the  $LC_{50}$  at a given target time, and the continuous curves correspond to the GUTS models). We can see on these figures that curves of the GUTS models do not start at day 0: at the beginning of the experiment, there were no dead snails and consequently it is not possible to estimate the  $LC_{50}$  at day 0. We notice again the similarity between GUTS-SD and GUTS-IT models. The target time  $LC_{50}$  estimates are close to GUTS ones, especially from a 35-days exposure duration. After 42 days of exposure, target time  $LC_{50}$  estimates do not change anymore, and the three types of estimates are very close both in median and in uncertainty.

## DISCUSSION

TKTD models are known to have several advantages over classical target time (TT) dose-response models mainly due to their mechanistic derivation. In this paper, we highlight the advantages of GUTS models to fit time- and concentration-dependent survival data in ecotoxicology. We show that GUTS models provide smaller 95% credible intervals and more conservative predictions of  $LC_x$  than classical TT analysis. We also show the impossibility to choose between GUTS-SD and GUTS-IT models based on either the estimated values of model parameters, their joint posterior distributions, their correlations and the time-course of the  $LC_{50}$  over 56 days, or based on dedicated tools for goodness-of-fit like DIC and cross-validation.

### ***TKTD models added-value***

The main difference between TKTD models and classical TT analyses is that TKTD models take into account both time and concentration simultaneously [9]. Therefore, all the collected data are used by TKTD models (and not only data at a given TT) [27], so that the dynamics of both the exposure and the effects can be better comprehended over time. Importantly, TKTD models enable extrapolating the results beyond the experimental conditions, as soon as the assumptions on the dynamic processes underlying the toxic response are made (in the scope of this paper it means choosing between GUTS-SD and GUTS-IT), as highlighted by Jager et al. [9]. For example, we could have modelled the time-course of survival for longer time, or it would have been possible to simulate time-varying concentration effects even though the concentration was held constant in experiments used to estimate model parameters [27]. More generally, environment displays plenty of situations that have not been tested in laboratories. In this context, obtaining robust predictions of adverse effects is a major goal for regulatory risk assessment. This goal is achievable with TKTD models.

Moreover, TKTD models add some mechanistic understanding in comparison with classical TT models, since they are built upon general biological processes [9]. In addition, GUTS parameters are time independent whereas they are time dependent in a TT model. It means that TKTD models give information on the biological processes that take place during the experiment through the biological significance and value of the estimated parameters [28]. Comparing and interpreting the results of different experiments and exposure scenarios is consequently easier and more meaningful with TKTD models [9]. For instance, we can see in Table S2 that the background mortality is the highest for Lab.05 with both GUTS-SD and GUTS-IT approaches. Having this knowledge on background mortality is helpful to disentangle between effects of the

chemical substance and effects of the experimental conditions. In addition, dominant rate constant  $k_d$  is related to the recovery of the organisms (its value enlightens about the speed of the underlying process) and the threshold concentration for effects (either represented by parameters  $z$  or  $\alpha$ ) is related to the no effect concentration on survival. This information is particularly interesting for the understanding of factors that affect sensitivity to a chemical substance in a given species and when comparing the sensitivity between species [28]. Getting similar threshold concentrations ( $z$  or  $\alpha$ ) with both models strengthens our confidence in the reliability of these parameters estimates. In our study, the dominant rate parameter related to the toxicokinetics part ( $k_d$ ) was greater for GUTS-SD compared to GUTS-IT. Such a trend was already observed in [29], so that we should cautiously consider the biological interpretation of its parameter value which may depend on several life-history traits (e.g., organism size). Accordingly, comparing both GUTS models has the potential to bring a better understanding of various possible effects and recovery mechanisms [28]. Therefore, this has the potential to bring more realism to the environmental risk assessment.

As regards the time-course of the  $LC_{50}$ , we can see on Figure 2 and Figure S4 in Supporting Information that GUTS-type approaches often provide more conservative predictions at intermediate exposure durations (between 14 and 38 days depending on the laboratory) than the TT analysis. Also, the 95% credible interval is often smaller with the GUTS models than with the TT analysis. These are important advantages of the GUTS models over the classical analyses, which appear particularly relevant in the field of environmental risk assessment. Indeed, environmental risk assessment could benefit from GUTS-based  $LC_{50}$  values because their estimation appears more precise due to the use of the entire data set. The possibility to compute  $LC_{50}$  values with their credible interval (that is the range of uncertainty) at any time and

concentration is also of particular interest, for example to make predictions in situations where data are lacking, or where new chemical exposure profiles need to be explored.

### ***GUTS-SD and GUTS-IT performance***

We highlighted in the previous section the advantages of the TKTD approaches. In the scope of this paper, we analysed the data with two of them: GUTS-SD and GUTS-IT. As illustrated, the time-course of the  $LC_{50}$  median values and of their 95% credible intervals, as well as the time-course of the predicted number of survivors and the posterior predictive checks are very similar for both models. Results from the cross-validation also reinforce this statement. It is consequently almost impossible to distinguish the two models based on a visual assessment of the fit quality as shown on Figures 1, 2, S1, S2 and S3. These results corroborate a similar statement by Ashauer et al., 2013, for pesticides and several species of fish [29]. The slight differences between the two models are more visible if we look at the estimated parameters. Table S2 and Figure S5 shows that the median of the background mortality  $h_b$  and of the threshold concentration (represented either by  $z$  or  $\alpha$ ) estimates as well as their 95% credible intervals are close when estimated with GUTS-SD and GUTS-IT (the difference is always below a factor 1.75 for  $h_b$  and 1.40 for the no-effect-concentration). The median of dominant rate constant  $k_d$  is however often much higher when estimated with GUTS-SD (until nearly 260 times higher for Lab.11). A sensitivity analysis was performed (see all Figures in the section “sensitivity analysis” of Supporting Information) to better understand the relative influence of each parameter on the shape of the predicted survival curve. In this purpose, three parameters over four were fixed and the remaining parameter was varied within  $\pm 50\%$ . This sensitivity analysis showed that  $k_d$  and  $z$  for GUTS-SD, or  $k_d$  and  $\alpha$  for GUTS-IT, are the parameters with the most important influence on the shape on the survival rate over time under our constant

exposure conditions. This result agrees with Ashauer et al., who also stressed that the sensitivity of modelled survival towards  $z$  and  $k_d$  differed a lot depending on the exposure pattern: for some of the exposure patterns this sensitivity could be high [29]. This is because  $k_d$  and  $z$  in the GUTS-SD model are correlated for some laboratories, and  $k_d$  and  $\alpha$  are strongly correlated for all the laboratories of the two ring-tests in the GUTS-IT model (as we can see on Figure S3 in Supporting Information). The consequence of this strong correlation is that changing one of the parameter values has an incidence on the other parameter value, as shown by the fact that some tested couples of  $(k_d, \alpha)$  values were not in the correlation plot line. The fact that the correlation between  $k_d$  and  $\alpha$  is strong for all the laboratories is probably inherent to the GUTS-IT model itself. This could lead us toward the use of the GUTS-SD approach as the preferred method. Nevertheless, if we look at the correlation plot of the GUTS-SD analysis on Figure S3, we can see that some posterior distributions (as for Lab.10) are bimodal. These bimodalities in the posterior distributions appear independently of the number of iterations per MCMC chain. They increase the uncertainty on outputs as the  $LC_{50}$  estimates. However, as we mentioned in the previous “TKTD models added-value” part, the uncertainty with TKTD models we tested can be quantified and still remains smaller in average than the uncertainty of the current classical TT analysis.

According to Ashauer et al. [30], the precision and accuracy of some parameters of the GUTS proper model (a combination of both GUTS-SD and GUTS-IT models), depend a lot on the number of individuals per experiment. In their study, the precision and accuracy of the killing rate  $k_k$  and of the median threshold  $z$  increased a lot when the initial number of individuals was fixed to 100 instead of 20 in every sample. As there were 30 snails per concentration for each laboratory in our data sets, we could assume that the uncertainty on the killing rate  $k_k$  for GUTS-

SD (especially high for laboratories in the validation ring-test) and of the median threshold ( $z$  in the GUTS-SD model or  $\alpha$  in the GUTS-IT model) could have been reduced by increasing the number of individuals in the experiments.

To summarize, it is difficult to select one of the reduced GUTS models presented in this paper as being overall better than the other, as Ashauer et al. [30] or Ducrot et al. [28] noticed already. Both models give close results and their performances depend on the criteria we want to focus on: the GUTS-IT model should allow to avoid bimodal posterior distributions, the GUTS-SD model should allow to avoid strong correlations between parameters, and the use of a TKTD approach provides the time-course of the  $LC_{50}$  with a reduced uncertainty by taking advantage of the whole set of raw data. Biological arguments as the species and/or the chemical substance could motivate the choice towards SD or IT. Nevertheless, based on our results for *L. stagnalis* exposed to Cd, there was no relevant biological reason to decide. While this could be a matter of concern from a statistical point of view, having two complementary models may be advantageous for environmental risk assessment. Indeed, fitting is fast enough to be performed with both models. Also, checking their goodness-of-fit can easily be handled based on tools as proposed in the present paper: the joint posterior distribution of parameters, the posterior predictive check, the deviance information criterion and the cross-validation. Then, if both SD and IT models fill all goodness-of-fit criteria, as was observed for most of our fits, this testifies the quality of the datasets, and therefore the implemented experimental protocols used to collect them. On the other hand, if both models show bad fit results, the experimental design or the sample size of the datasets could be questioned. In between, if one model appears better than the other, the choice of the most appropriate one should depend on the question at hand.

426 Based on the data sets we studied, our results highlight the added-value of TKTD models, which  
427 take into account both time and concentration simultaneously, and which are based on the  
428 internal processes that take place in response to the exposure to a chemical substance.  
429 Considering our results, it was difficult to define one criteria which could help choosing between  
430 GUTS-SD or GUTS-IT. Indeed, both models fitted pretty well the observations and each of them  
431 had pros and cons as regards the parameter estimates. Consequently, picking one model more  
432 than the other depends on the specific requirements and no general rule can be dictated.

## ASSOCIATED CONTENT

**Supporting Information.** The .PDF file provides all complementary fitting results for all laboratories of the two ring-tests, as well as the sensitivity analysis. This information is available free of charge via the Internet at <http://pubs.acs.org>. Raw data are available upon request from the corresponding author.

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. ‡ These authors contributed equally.

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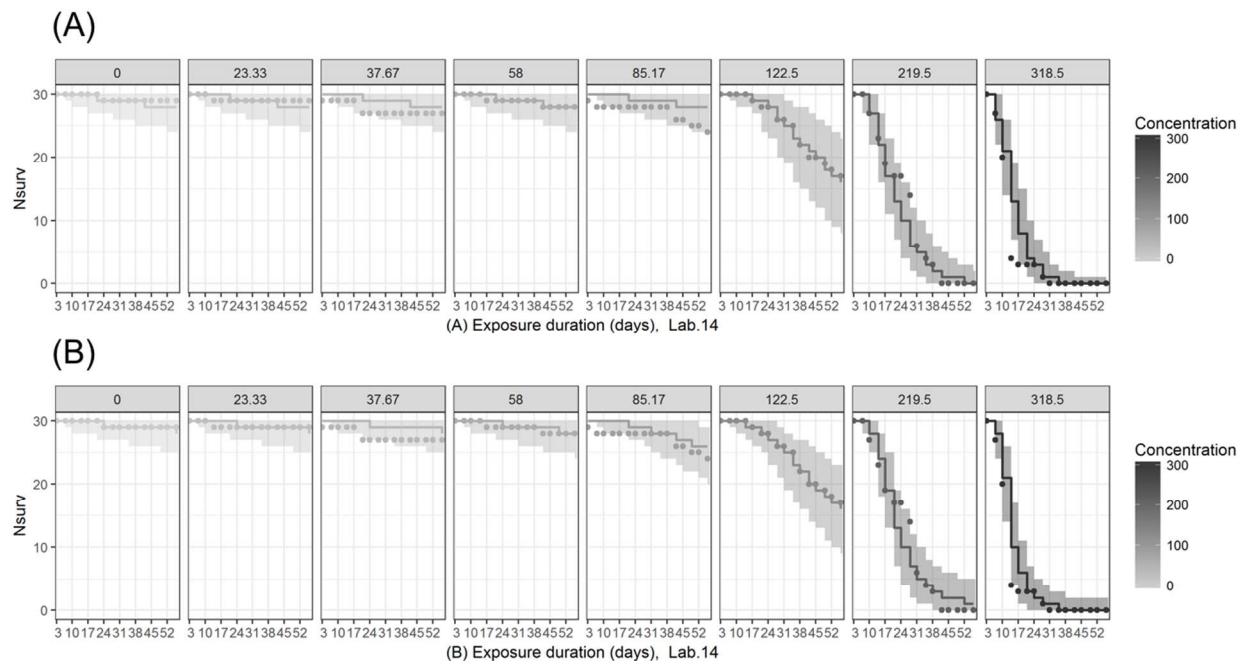
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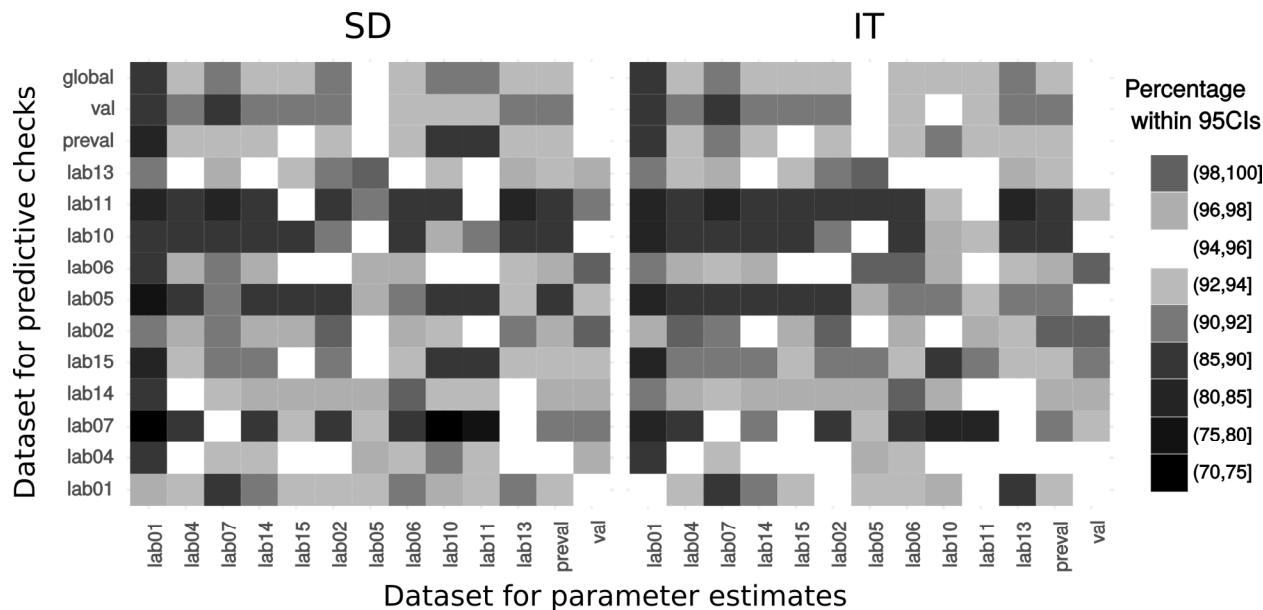
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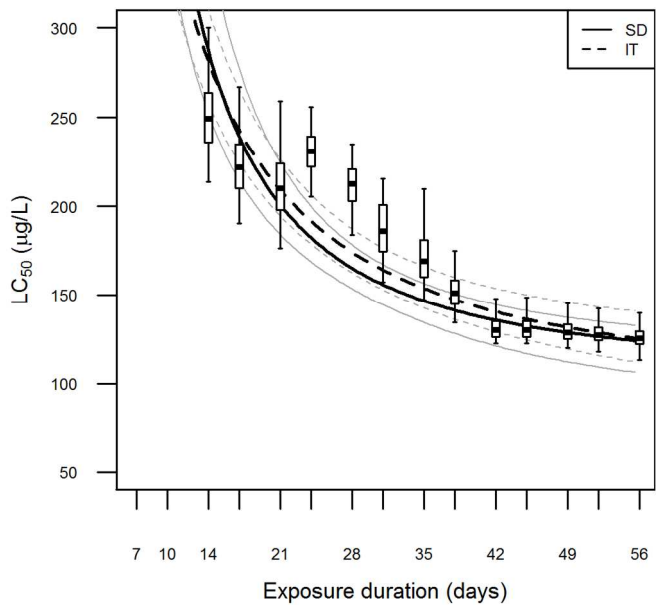
**Figure 1.** Observed numbers of survivors (dots) and simulated numbers of survivors associated to the 95% credible band: (A) GUTS-SD model and (B) GUTS-IT model for Lab.14.



**Figure 2.** Average percentage of observed points in the 95% credible intervals of prediction.

Parameters are estimated from data sets of the  $x$ -axis, and 95% credible intervals of predictions

are compared to data sets of the y-axis. Grey levels indicate the percentage of data lying within the 95% credible intervals.



**Figure 3.** Time-course of the LC<sub>50</sub> (black curves) and their 95% credible intervals (grey curves) as estimated at target time (boxplots), with GUTS-SD (continuous lines) and GUTS-IT (dashed lines) models for Lab.14. The observed number of death before day 14 was too low to perform target-time analyses and estimate LC<sub>50</sub> values.

**Table 1.** Parameters and symbols used in the GUTS-SD and GUTS-IT models. Letter  $d$  stands for the time unit in days, while (-) stands for dimensionless.

| Parameter                                          | Symbol   | Unit                  | Model     |
|----------------------------------------------------|----------|-----------------------|-----------|
| Dominant rate constant                             | $k_d$    | $d^{-1}$              | IT and SD |
| Background mortality                               | $h_b$    | $d^{-1}$              | IT and SD |
| Threshold for effects (or no effect concentration) | $z$      | $\mu g.L^{-1}$        | SD        |
| Killing rate                                       | $k_k$    | $L.\mu g^{-1}.d^{-1}$ | SD        |
| Median of the threshold distribution               | $\alpha$ | $\mu g.L^{-1}$        | IT        |
| Shape of the threshold distribution                | $\beta$  | (-)                   | IT        |